

IS6400 Business Data Analytics

Dr. LIU Junming

City University of Hong Kong

Lecture 1

Reading Material:

☒ Course Information

☐ Introduction

☐ Course Overview

☐ Platform Introduction

Instructor Information

❖ Instructor:

Junming Liu, Ph.D. AC3 6-277 (3442-9323)

junmiliu@cityu.edu.hk

❖ Co-Instructor and TA:

∞ SHUAI Yanqin, yshuai8-c@my.cityu.edu.hk

∞ QIN Yaxue, yaxueqin2-c@my.cityu.edu.hk

∞ Yu Shujing, shujingyu2-c@my.cityu.edu.hk

Office Hour: Mon. 12:00-14:00, by email appointment

Class Policy

❖ No Cell Phone in classroom

- Pay mutual respect to each other
- Do not disturb your fellow classmates



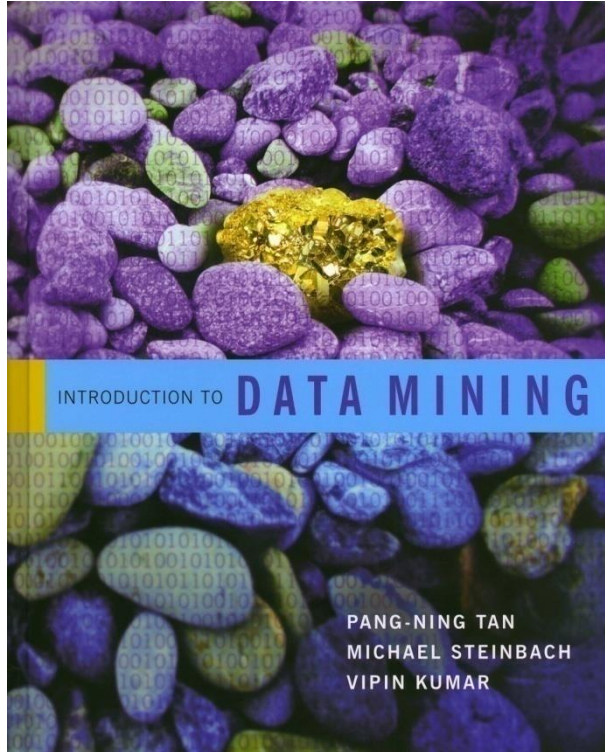
❖ Attend the classes punctually

❖ Academic Honesty

- Plagiarism is a SERIOUS offense at CityU
- Don't do it. If you're in doubt, discuss with your lecturer or tutor

Required Textbook

DM (modeling part)

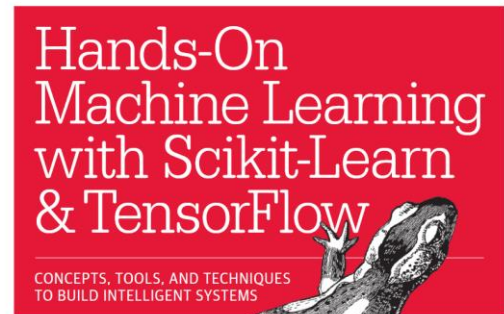


Introduction to Data Mining

Pang-Ning Tan, Michael Steinbach, Vipin Kumar,
ISBN: 0-321-32136-7, 2005. [eBook Link](#)

HM (source codes)

O'REILLY



Aurélien Géron

Hands-On Machine Learning with Scikit-Learn & TensorFlow

Course Materials

- ❖ Platform: **Anaconda** – The World's Most Popular Python/R Data Science Platform
 - ❖ <https://www.anaconda.com/download>
- ❖ Library: Data Mining techniques using Python packages
 - <http://scikit-learn.org/stable/>
 - http://scikit-learn.org/stable/auto_examples/index.html
- ❖ Library: TensorFlow
 - <https://www.tensorflow.org/>

Tentative Teaching Schedule

COURSE SCHEDULE (*TENTATIVE*)

Week#	Lecture	Hands-on Tutorial	Tutor
1	Course Introduction	Software installation	<u>Yaxue Qin</u>
2	Predictive Analytics: Multivariate regression	Regression Analysis	<u>Yaxue Qin</u>
3	Feature Selection and Dimension Reduction	Principle Component Analysis	<u>Yaxue Qin</u>
4	Bisecting <u>KMeans</u> , DBSCAN	Clustering Methods	<u>Yaxue Qin</u>
5	Supervised Learning: Decision Trees, Ensemble Learning	Classification Methods	<u>Yaxue Qin</u>
6	Project Briefing		
7	Time series analysis (1)	Holt's Model	<u>Yanqin Shuai</u>
8	Time series analysis (2)	Exponential smoothing	<u>Yanqin Shuai</u>
9	Deep Learning: ANN and RNN	Nonlinear Time Series	<u>Yanqin Shuai</u>
10	Deep Learning: Image Mining	Image Recognition	<u>Yanqin Shuai</u>
11	Model Assessment Competition: GenAI for BDA	Model Selection	<u>Yanqin Shuai</u>
12&13	Group Project Presentation		

Course Intended Learning Outcome

- ❖ **Understand** the importance of data analytics for business
- ❖ **Learn** the Advanced concepts and techniques of Machine Learning and Deep Learning
- ❖ **Apply** learned techniques for Real-world problems
- ❖ **Identify** appropriate tools and techniques for innovative business solutions

Show something on your cv!

Assessment Criteria

- ❖ Participation (5 %)
 - Class attendance and participation, in-class quiz, etc.
- ❖ Group Project (30%)
 - Project Briefing (5%)
 - Presentation (15%)
 - Final report (10%)
- ❖ Individual Assignments (25 %)
 - 5~10 assignments due next WEEK
- ❖ Final Exam (40%) – 2hour (open/closed book)

Attendance

Regular attendance is compulsory. You are not allowed to check your emails, access Web sites not related to the course or work on something that is beyond the scope of this course during the class time.

If you want to finish your assignment quickly?

If you want to complete your project?

If you want to get some extra credits?

Assignments

You may have discussions with your class members, but you must submit your own work. Please be sure to keep a copy of the assignment by yourself in case that there is any problem with your hand-in or you have to use it later this semester.

Submit before the beginning of the class on the specified due day.

Late submissions will be accepted. (-20 points per day overdue)

Course Project

- A course project using any data analytics models using real word data.
- Max 6 students in a group.
- Dataset Available:
 - Kaggle: <https://www.kaggle.com/datasets>
 - Kaggle: <https://www.kaggle.com/competitions>
 - UCI: <https://archive.ics.uci.edu/ml/index.php>
 - Google Dataset:
<https://datasetsearch.research.google.com/>

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- ☐ Course Information
- ☒ **Introduction**
- ☐ Course Overview
- ☐ Platform Introduction

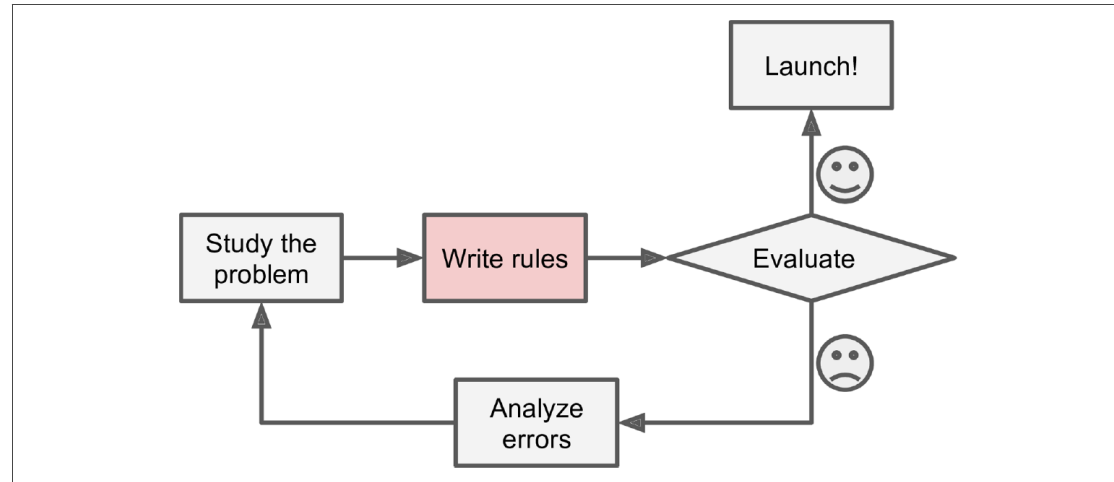
What is Data Analytics?

- **Tell computers to learn from Data**
 - Data Analytics is the process of automatically **discovering useful information in large data repositories**.
 - It is a **technology** that blends traditional data analysis methods with sophisticated algorithms for **processing large volumes of data**.
 - It has also opened exciting opportunities for **exploring and analyzing new types of data** and for analyzing old types of data in new ways.

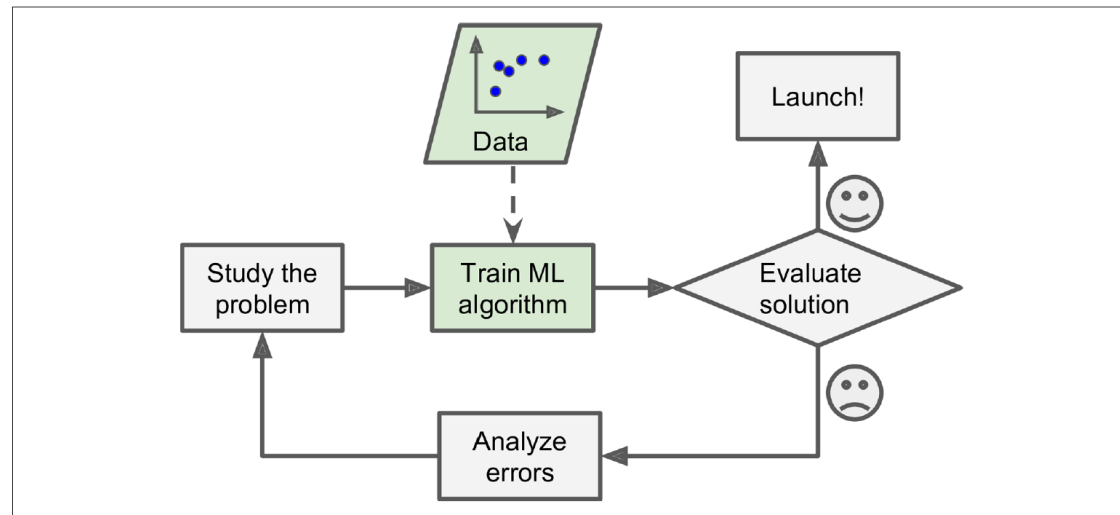
Machine Learning for DA?

Tell computers to learn from Data like a human

Human Learning

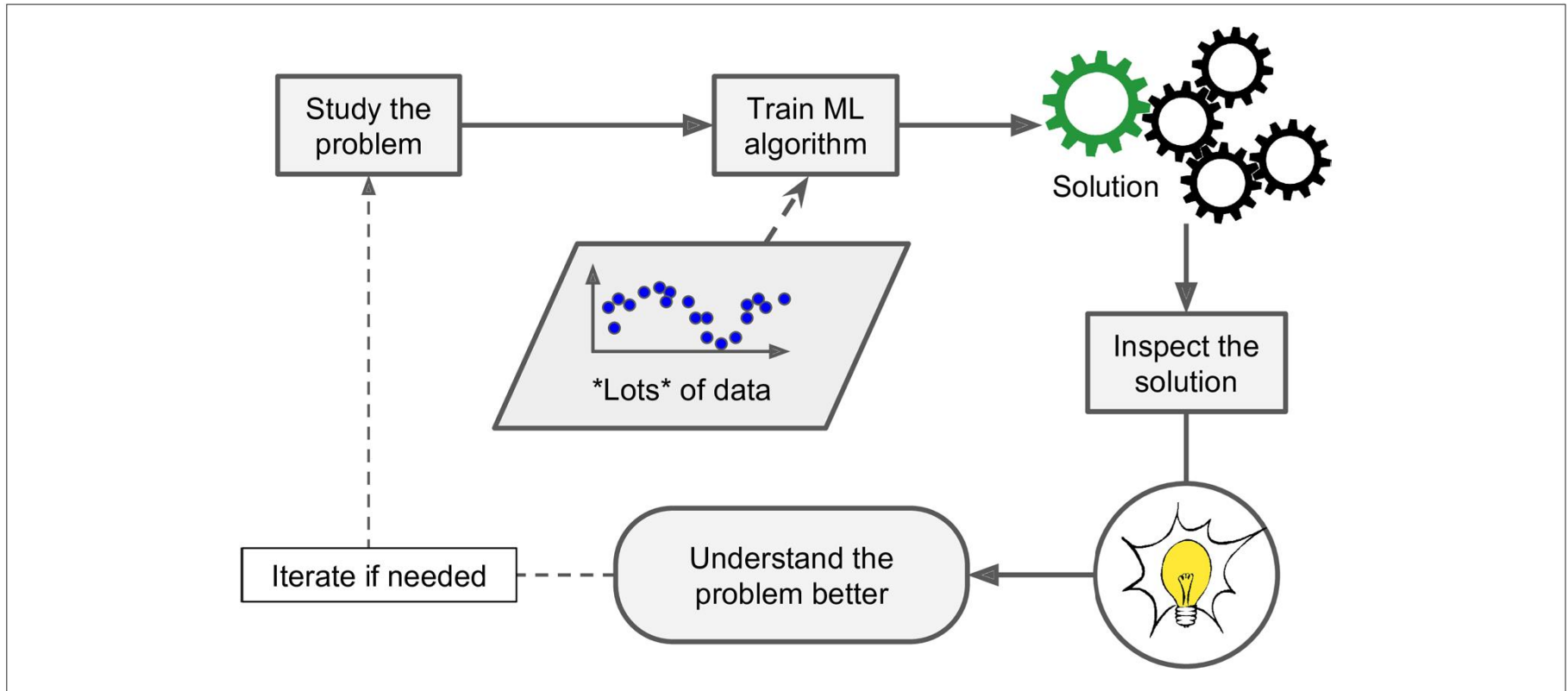


Machine Learning



Why use Machine Learning?

Help a human learn with a machine



Real-world Applications

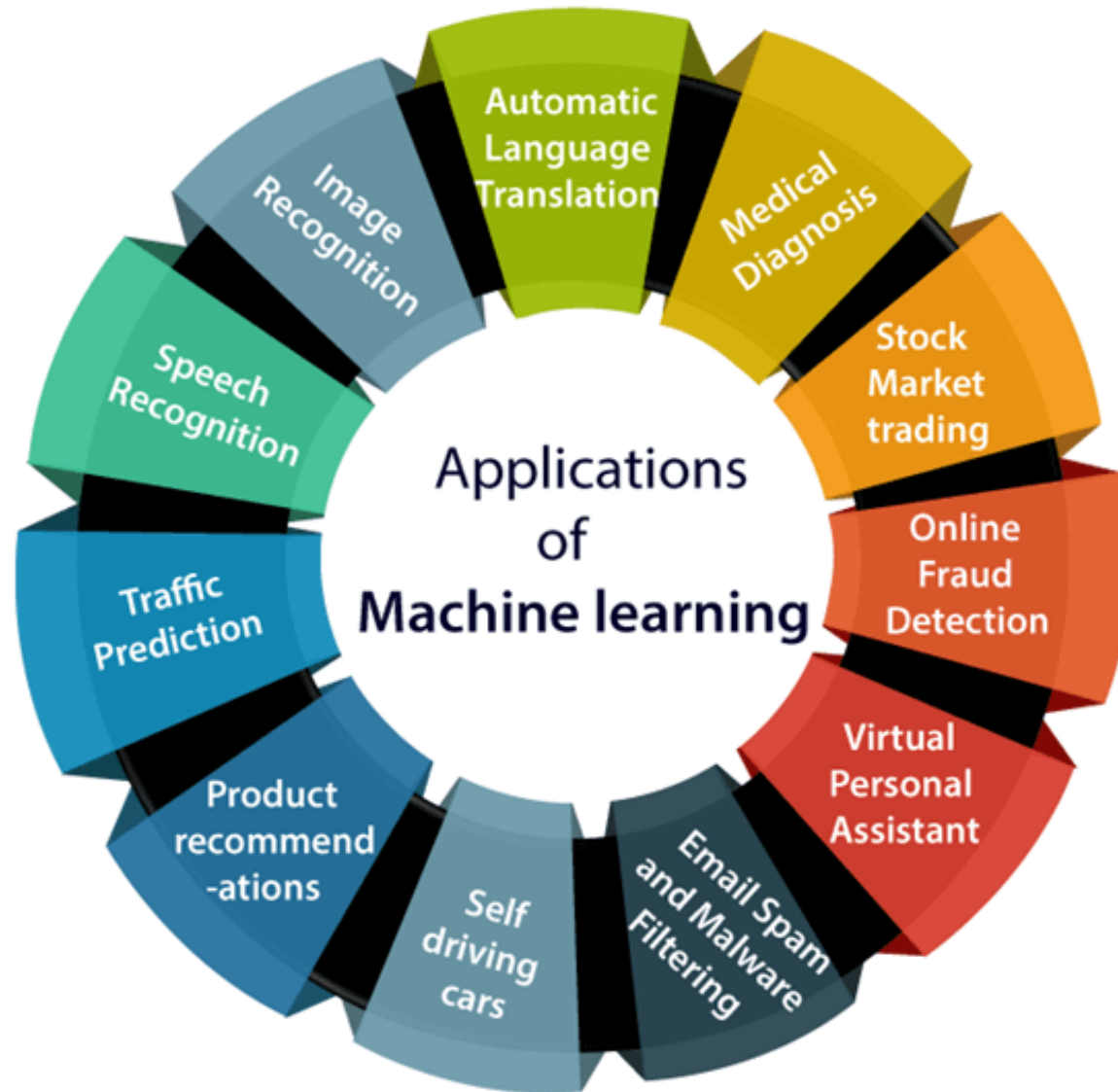


Image recognition – Example

❖ Image Recognition and Artificial Intelligence



Stock Market – Example

❖ Stock price forecasting <http://theoreti.ca/?p=5032>

Apple Inc. (AAPL)

Add to Watchlist

Add to Portfolio

519.80 \$1.34 (-0.26%)

11/24/2013 8:49pm EST

NYSEMKT - Disclaimer
Currency in USD

Volume: 7,979,492

Opening Price: 519.6

52 Week: 385.1 - 594.59

Market Cap: 467.7B

Social Sentiment (10/16/2013)



Strong Buy: 64%
Buy: 15%
Hold: 9%
Sell: 3%
Strong Sell: 9%

Chatter Volume

5 Day

2%

Today

2%



Chart

Alerts

Stock Chatter

News

Show Real-Time

\$501.11

Zoom

1m

3m

6m

YTD

1y

2y

5y

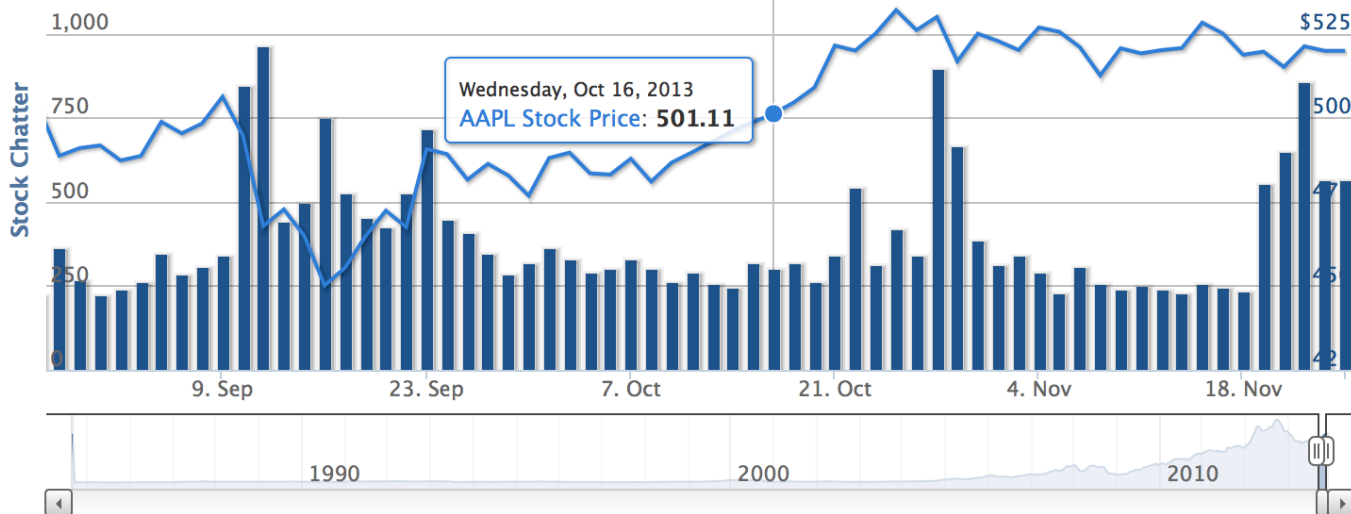
All

\$31.00 (6%) From

Aug 26, 2013

To

Nov 24, 2013

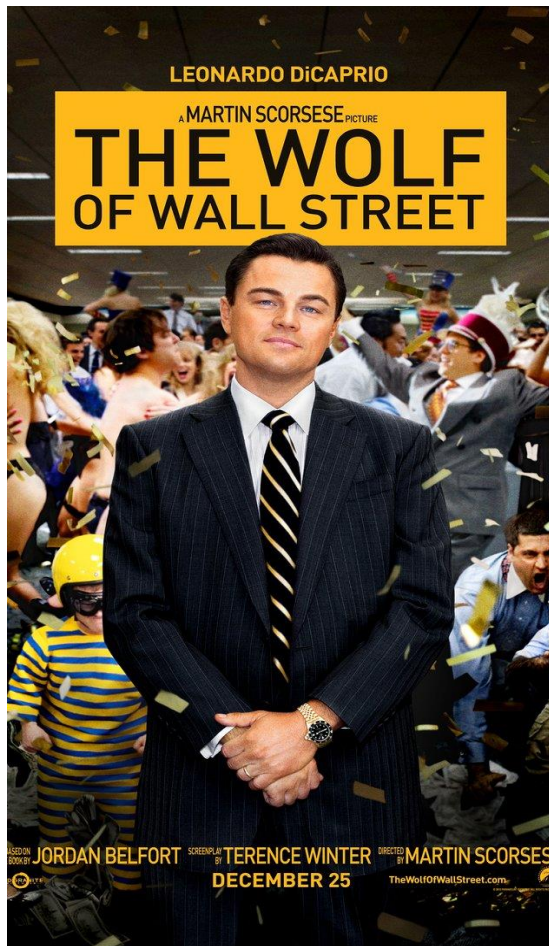


Clustering – Social Network Segmentation



Why Machine Learning for DA?

The Age of Big Data need machine to process data and support decision making



Life in 1990's



Life in 2010's

My Business Applications

Intelligent Production, Replenishment and Transshipment System for Retailers (perishable product and fast fashion)



Unbalanced consumption
among products in-store



Redistribute Stocks

Product List:
P1, P2, P3,...



Unbalanced consumption
among retailers

Representative Business Clients:



Fast Fashion
4300+ Retail Stores



Dairy Company
1800+ Retail Stores



Sport Fashion
1200+ Retail Stores

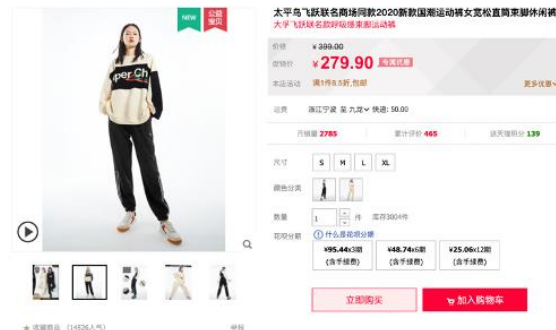
My Ongoing Applications

Use social media to shape fashion trend



Social
Media
Activity

Fast Fashion
Top Selling



Business Applications

❖ Pricing

- setting prices for consumer and industrial goods, government contracts, and maintenance contracts

❖ Customer segmentation

- identifying and targeting key customer groups in retail, insurance, and credit card industries

❖ Merchandising

- determining brands to buy, quantities, and allocations

❖ Location

- finding the best location for bank branches and ATMs, or where to service industrial equipment

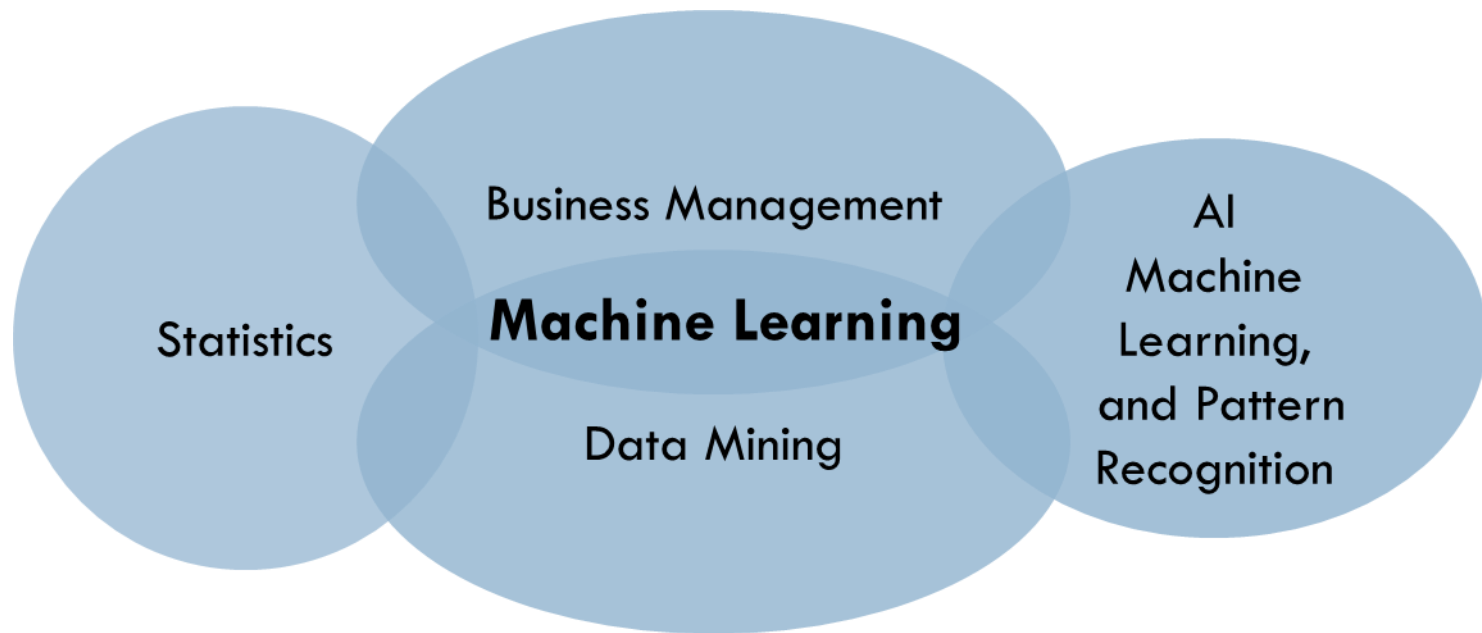
❖ Social Media

- understand trends and customer perceptions; assist marketing managers and product designers

Machine Learning Applications

- ❖ Market analysis
- ❖ Risk analysis and management
- ❖ Fraud detection and detection of unusual patterns (outliers)
- ❖ Text mining (news group, email, documents) and Web mining
- ❖ Stream data mining

Interdisciplinary Study



You Play in everyone's sandbox

Automotive
Education
Energy
Insurance
Transportation & Logistics
Information Technology
Entertainment
Government
Consulting
Telecommunications
Electronics
Food Processing
Retail Trade
Banking
Manufacturing

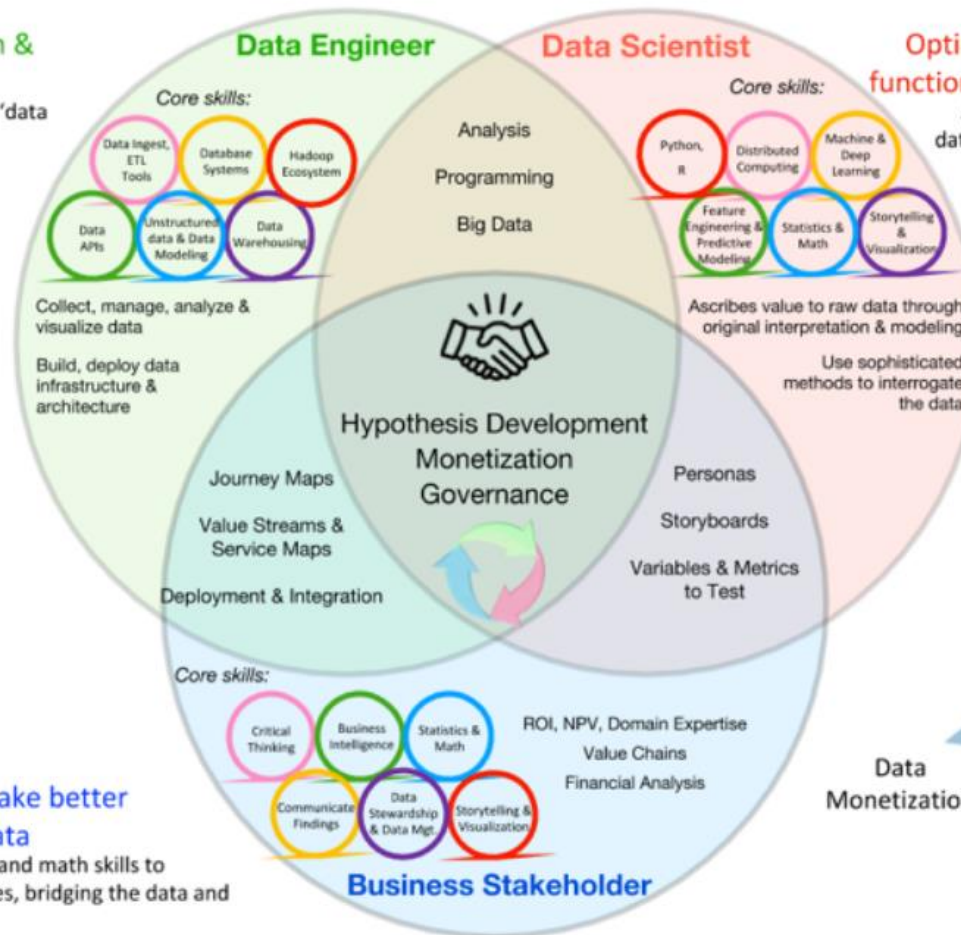
The best thing about analytics is that it lets you play in everyone's sandbox

Jobs in BDA

Data Science Roles & How They Interact

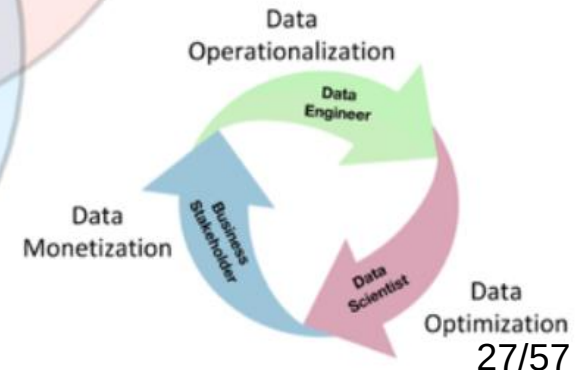
Enable data access & utilization & enable value capture

Builds and supports the infrastructure or 'data pipe' and all associated SW engineering infrastructure tasks.



Help the business make better decisions through data

Blend of business, analytic and math skills to explore and solve challenges, bridging the data and business communities.



Career Future

❑ Shortage:

USA alone faces a shortage of 140,000 to 190,000 people with deep analytical skills

❑ Shortage:

1.5 million managers to analyze big data and make decisions based on their findings.

❑ Data Scientist: The Sexiest Job of the 21st Century – Harvard Business Review

Just to find a good JOB!

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Lecture 1

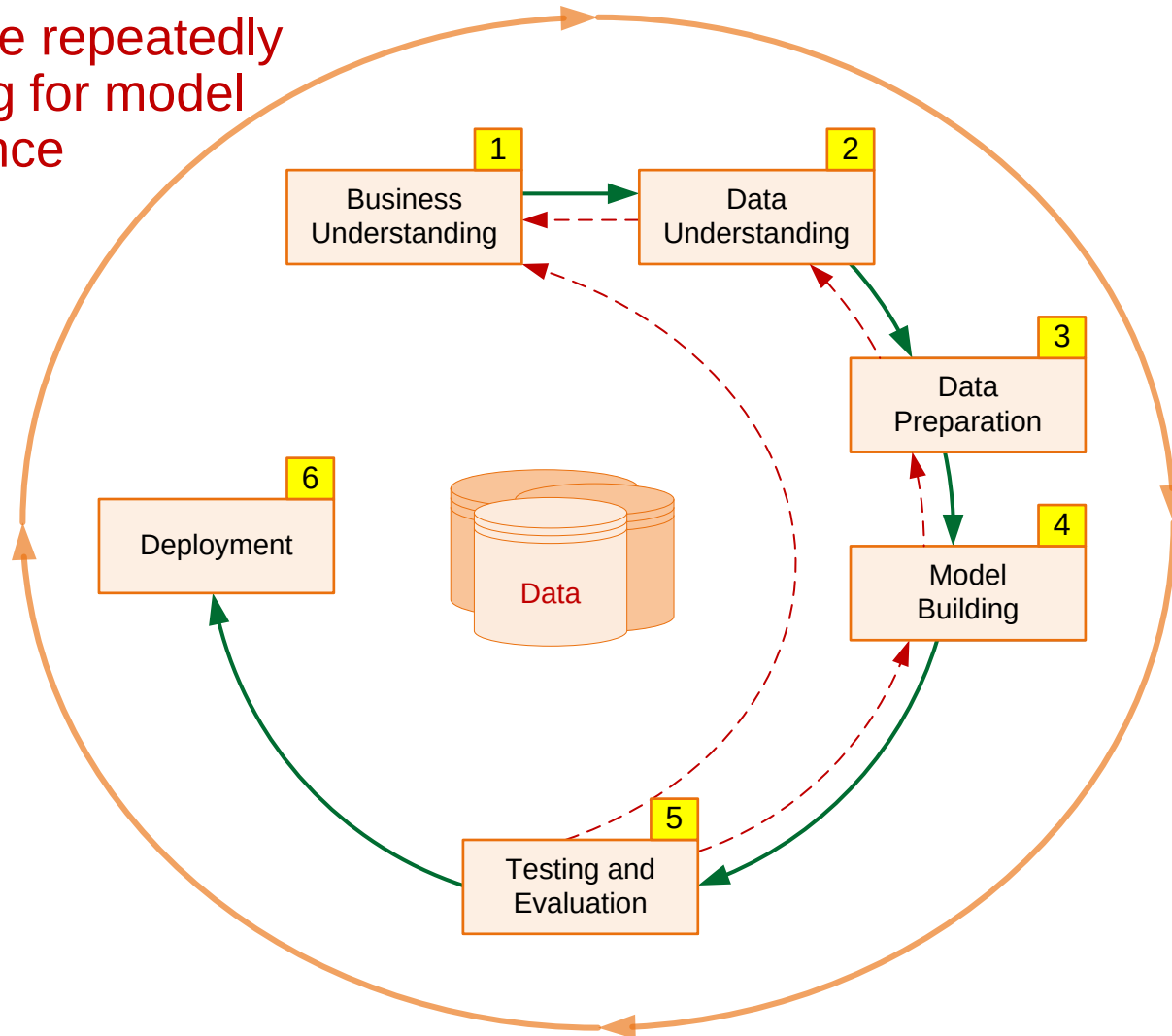
- ☐ Course Information
- ☐ Introduction
- ☒ **Course Overview**
- ☐ Platform Introduction

Reading Material:

- ☒ **DM Chapter 1**

BDA Process

Steps can be repeatedly
backtracking for model
& performance
refinement



BDA Process

- ❖ **C**ross **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining
- ❖ Proposed in 1990s by a European consortium
- ❖ Composed of six consecutive phases:
 - Step 1: Business Understanding
 - Step 2: Data Understanding
 - Step 3: Data Preparation
 - Step 4: Model Building
 - Step 5: Testing and Evaluation
 - Step 6: Deployment

Accounts for
~80% of total
project time

Your group project may follow this process (except deployment if not applicable)

Course Structure Overview

Advanced Machine Learning Techniques

Predictive Analytics

Week 2

Feature Engineering

Week 3

Clustering Models

Week 4

Prediction Models

Week 5-8

Neural Networks and Deep Learning

Artificial Neural Network

Week 9

Recurrent NN

Week 9

Image Mining

Week 10

GenAI and LLM

Week 11

End-to-end Business Applications

Feature Engineering



Feature is the element that you use to describe an object



Original photographer unknown

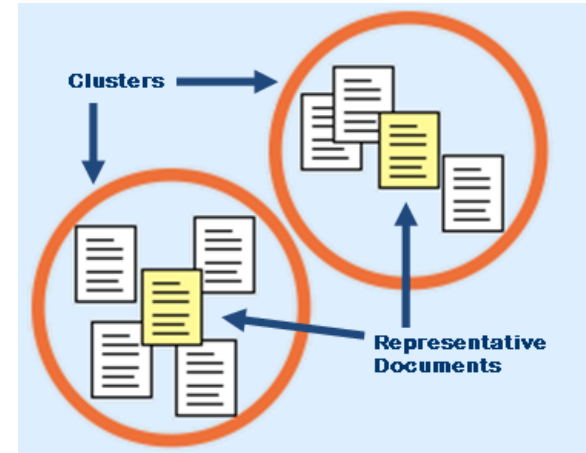
See also www.cs.gmu.edu/~jessica/DimReducDanger.htm

(c) eamonn keogh

Clustering

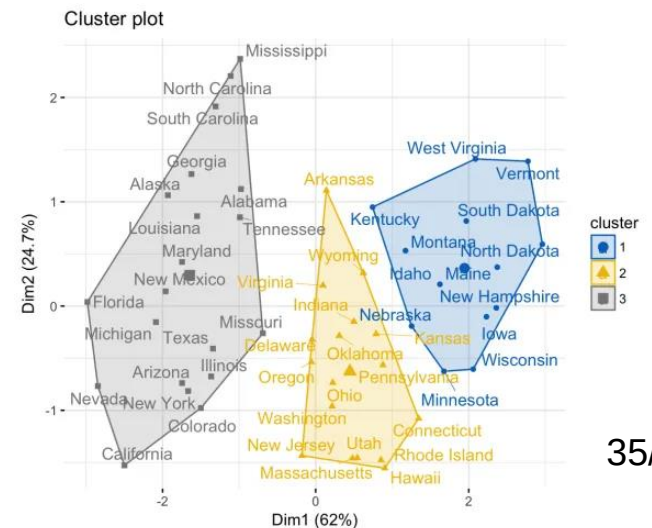
Understanding

- Group related documents for browsing, group genes and proteins that have similar functionality, or group stocks with similar price fluctuations



Summarization

- Reduce the size of large data sets



Clustering

❖ Clustering Algorithms

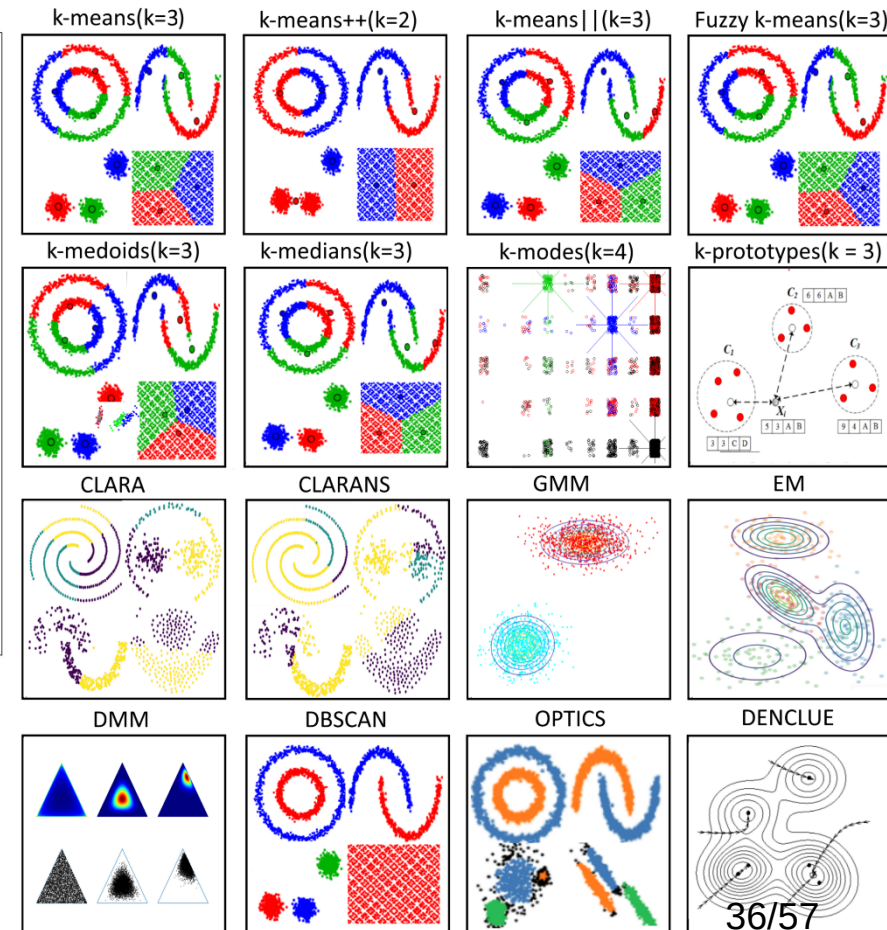
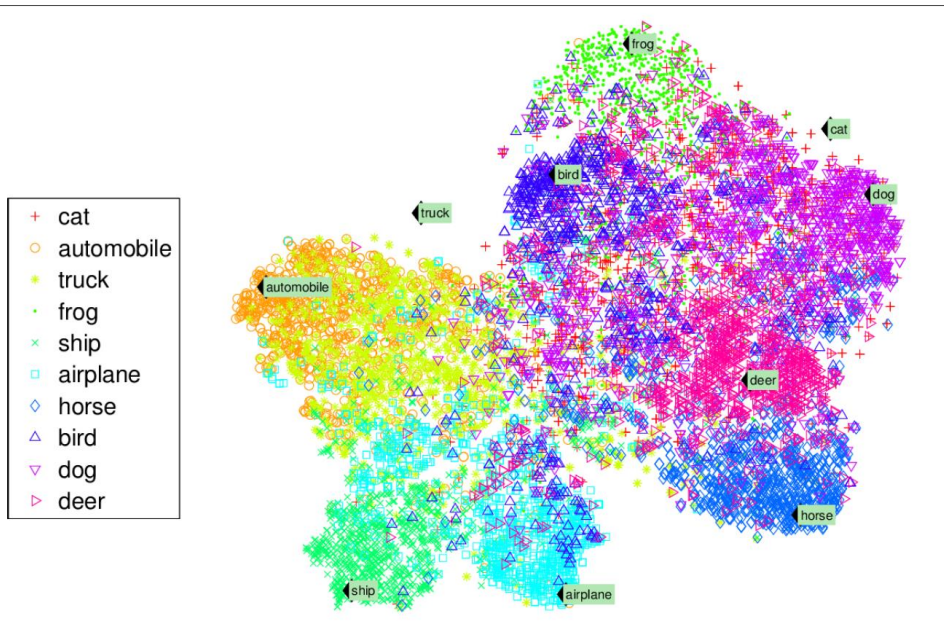
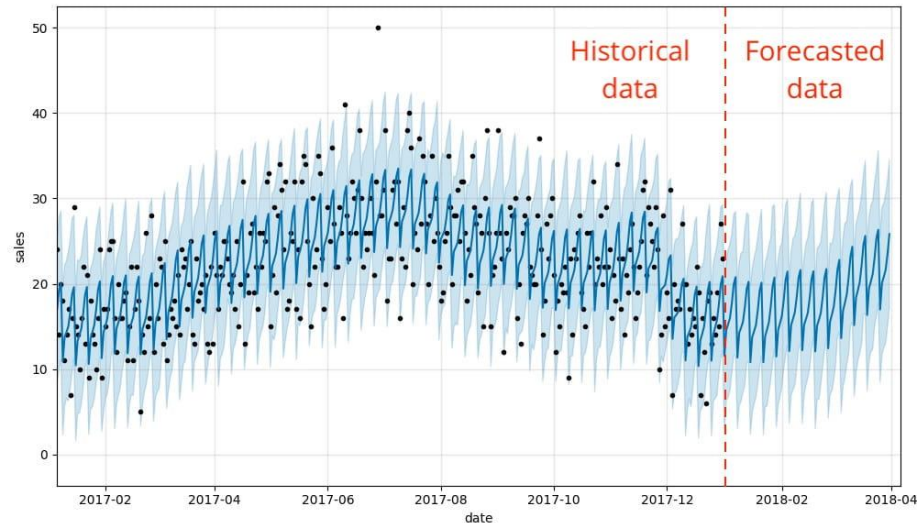
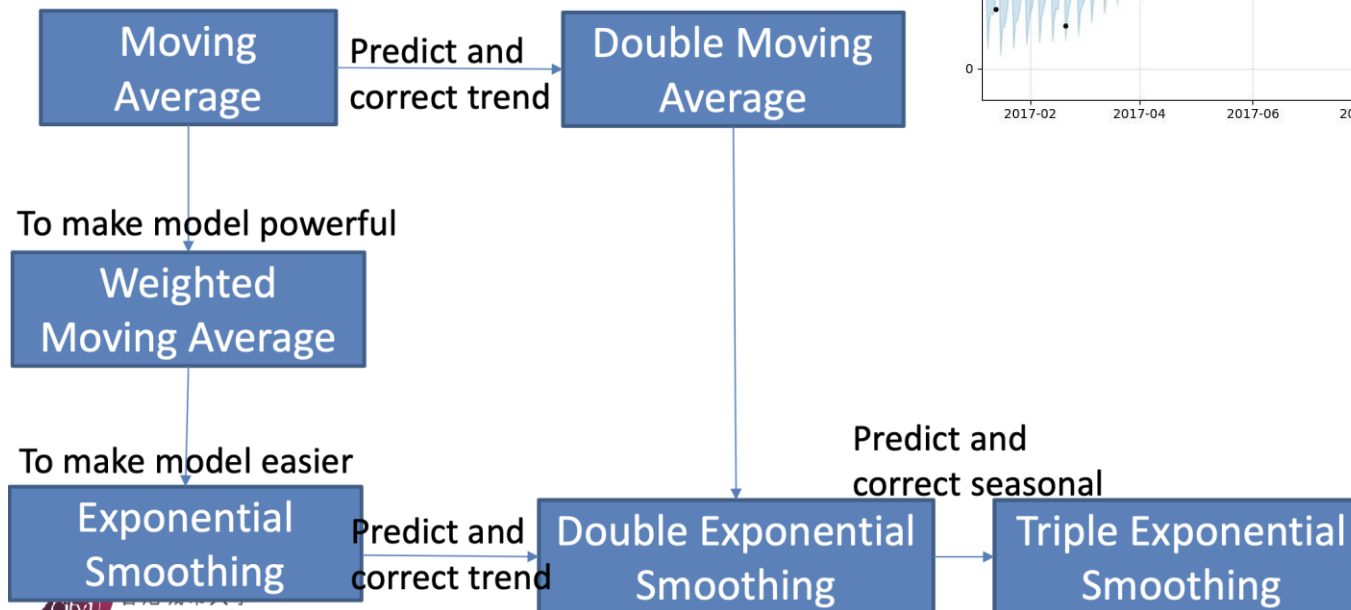


Figure 1-9. Example of a t-SNE visualization highlighting semantic clusters³

Predictive Analytics

Time series forecasting



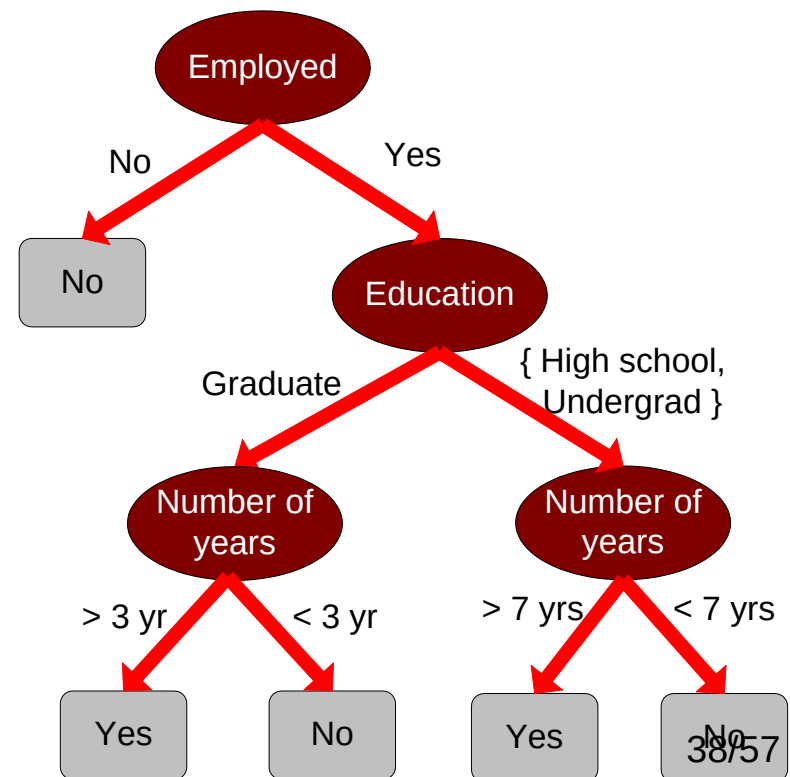
Classification

- ❖ Find a model for class attribute as a function of the values of other attributes

Class

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

Model for predicting credit worthiness



Advanced Classification

❖ Ensemble Learning to solve the same problem

Class

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

Model for predicting credit worthiness

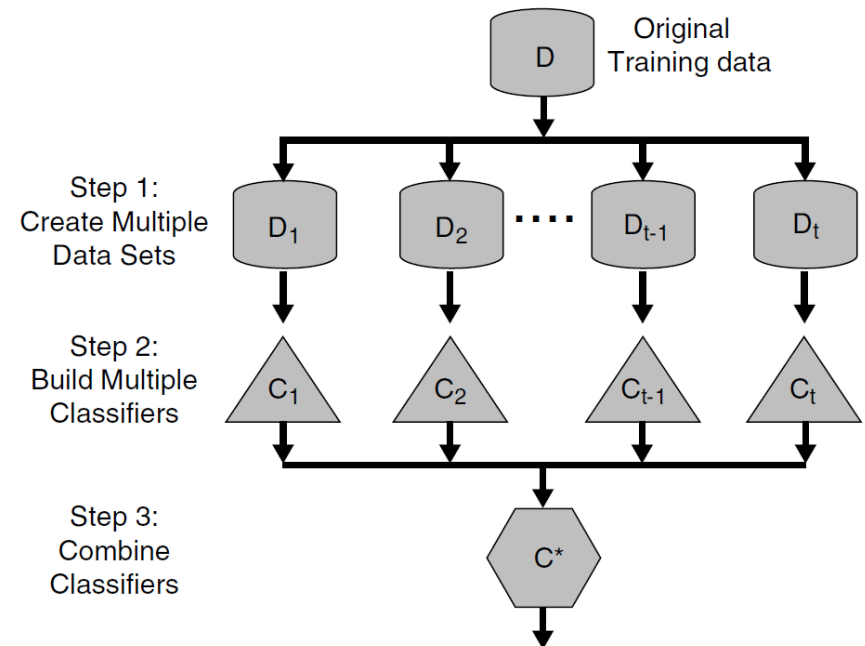


Figure 5.31. A logical view of the ensemble learning method.

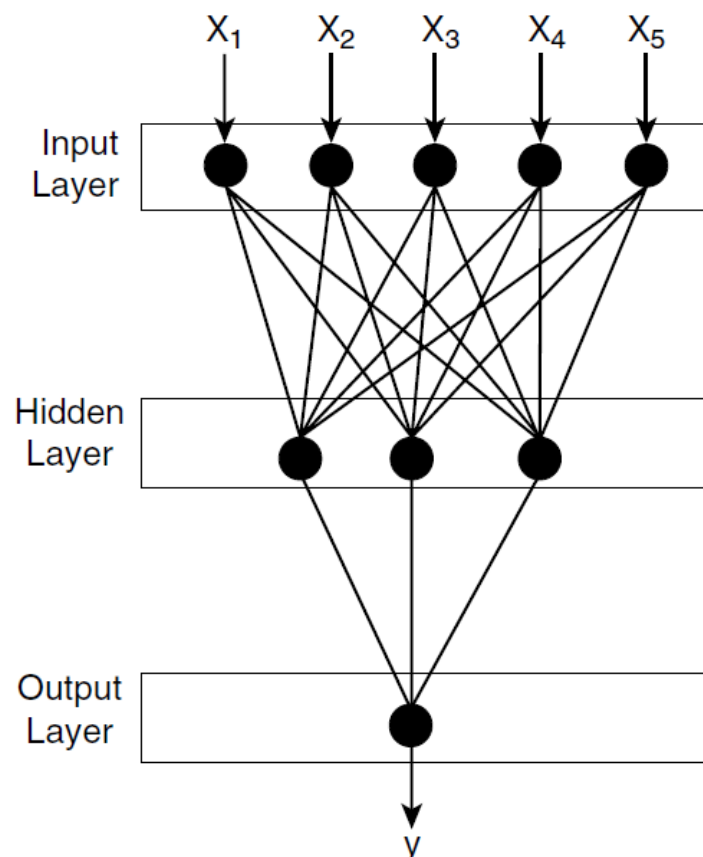
Deep Learning

❖ Neural Network

Class

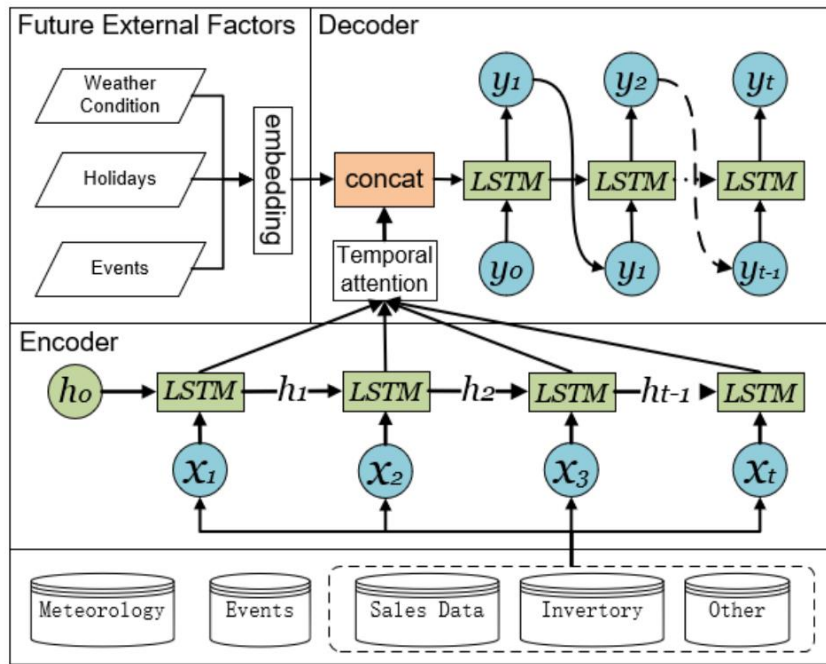
<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
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4	Yes	High School	10	Yes
...	...	...	...	...

Model for predicting credit worthiness

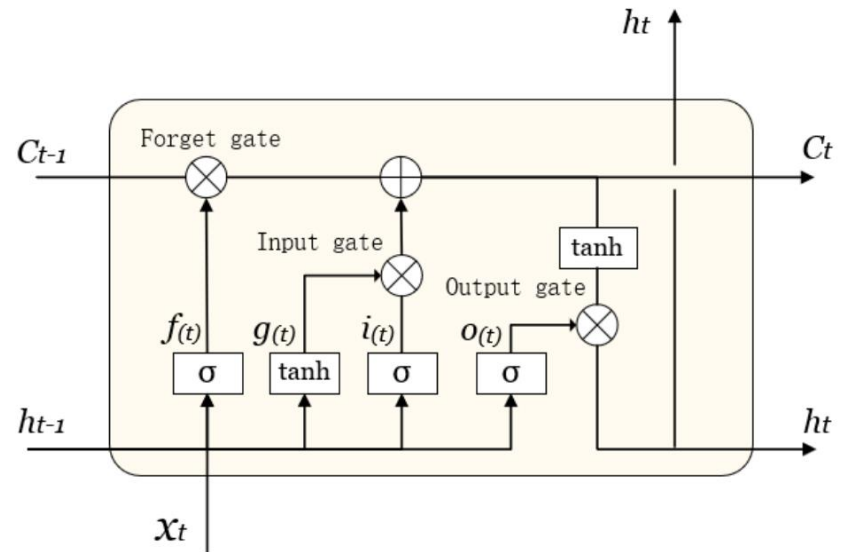


Neural Network

❖ Sequential Model



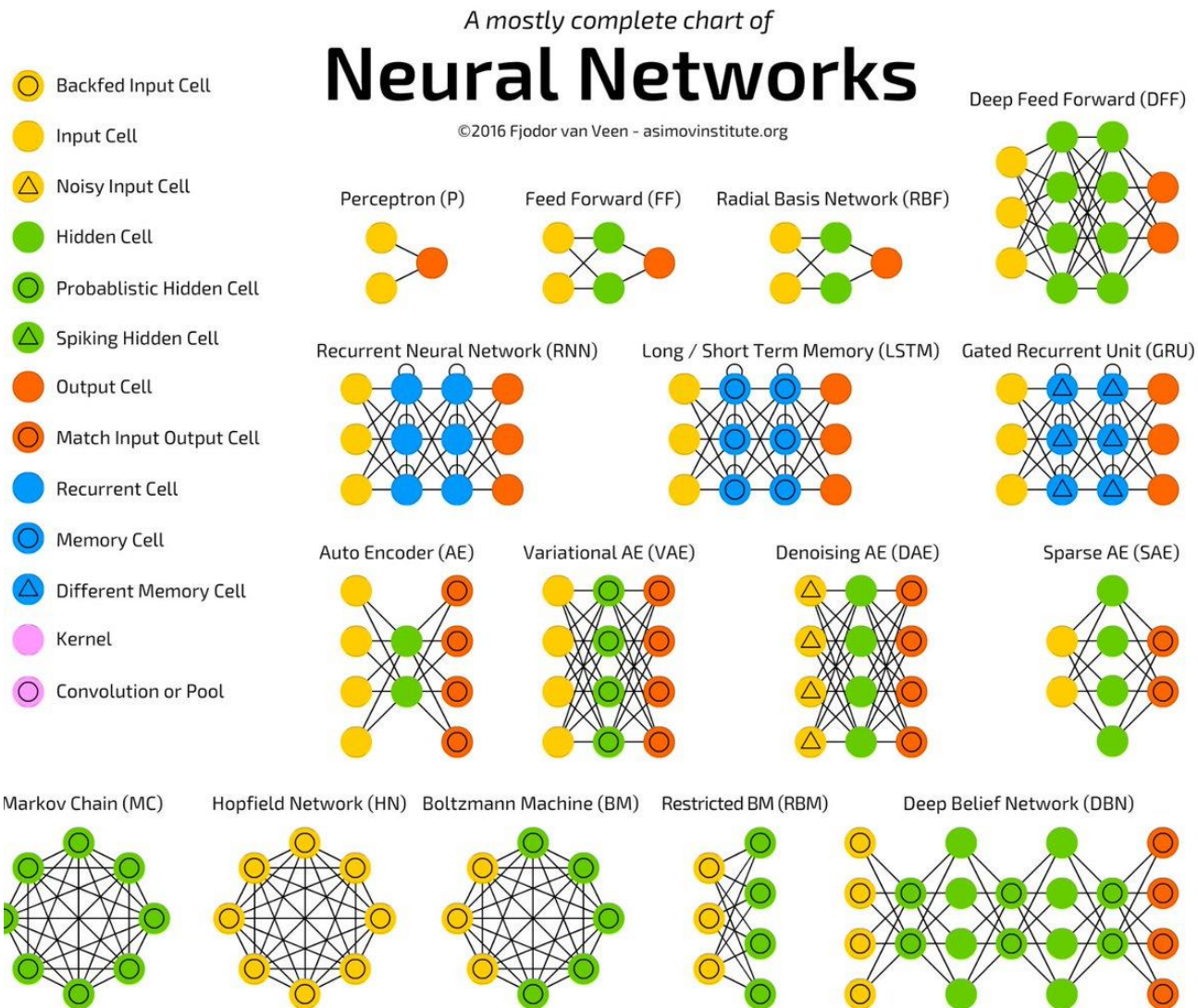
(a) Sequential Model



(b) LSTM Cell

Figure 2: Framework of Deep Sequential Model for Component Consumption Prediction

Neural Network Class



Deep Learning and GenAI

Deep Learning
(Methodological/technical framework)

Prediction, Classification

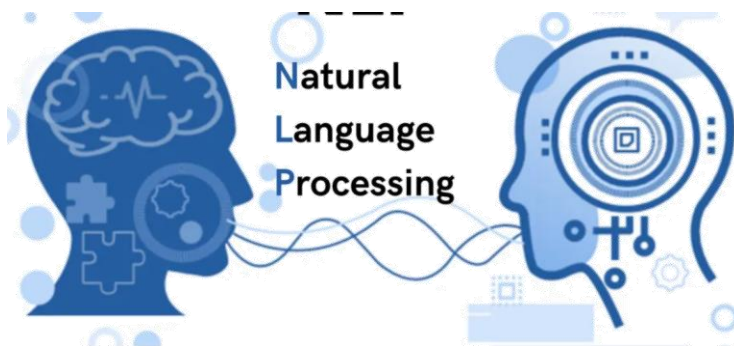


Generative AI

Extend to generate new content:
text, image, video

Application: **Natural Language Processing (NLP)**

Enabling computers to understand, process and generate natural language.



Large Language Model

❖ Language Model

Models capable of generating, understanding, or manipulating text or speech in natural language

- ❧ Neural network-based models

- ❧ Transformer models

- ❖ Attention: Instead of reading word by word, it can examine multiple words at once and understand their **relationships**

❖ Large Language Model

- ❧ Large model using Transformer

- ❧ Trained on large-scale text corpora

- ❧ e.g. Generative pre-trained transformer (GPT)

LLM Applications



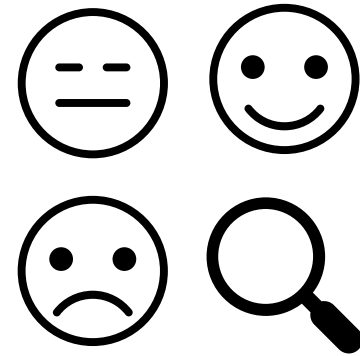
Chatbot

Text

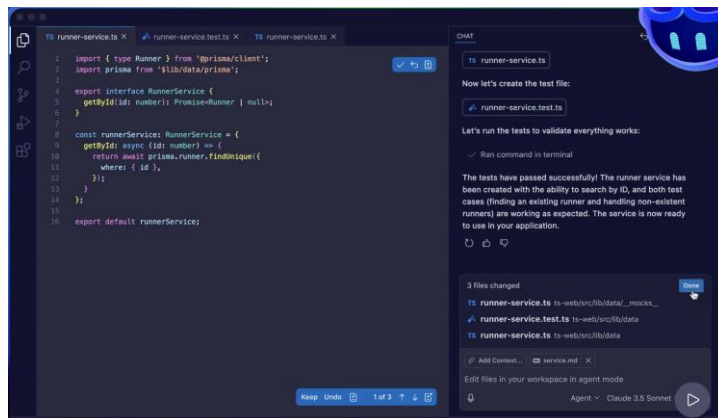
Image

Video

Content generation



Sentiment analysis



Code development

- Translation
- Recommendation
- ...

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- ☒ HM Chapter 1

Anacoda - Jupyternotebook

❖ <https://www.anaconda.com/>



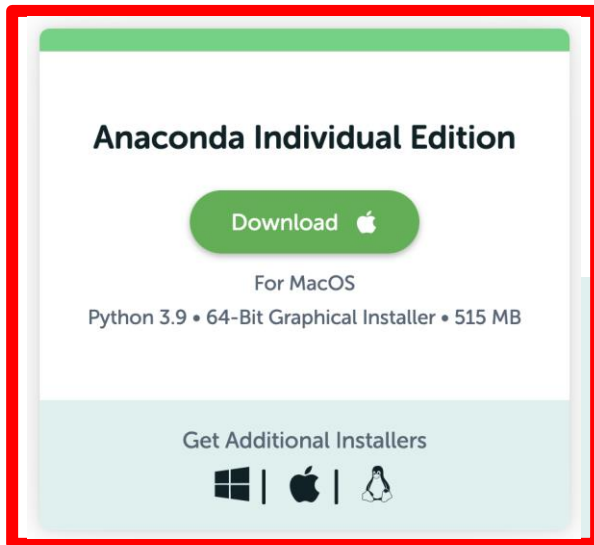
Anaconda was named a May 2019 Gartner Peer Insights Customers' Choice for Data Science and Machine Learning Platforms.



Reviews from enterprise peers—verified by Gartner

Anacoda - Jupyternotebook

❖ <https://www.anaconda.com/products/individual>



Anaconda Installers

Windows

Python 3.9

64-Bit Graphical Installer (510 MB)

32-Bit Graphical Installer (404 MB)

MacOS

Python 3.9

64-Bit Graphical Installer (515 MB)

64-Bit Command Line Installer (508 MB)

Linux

Python 3.9

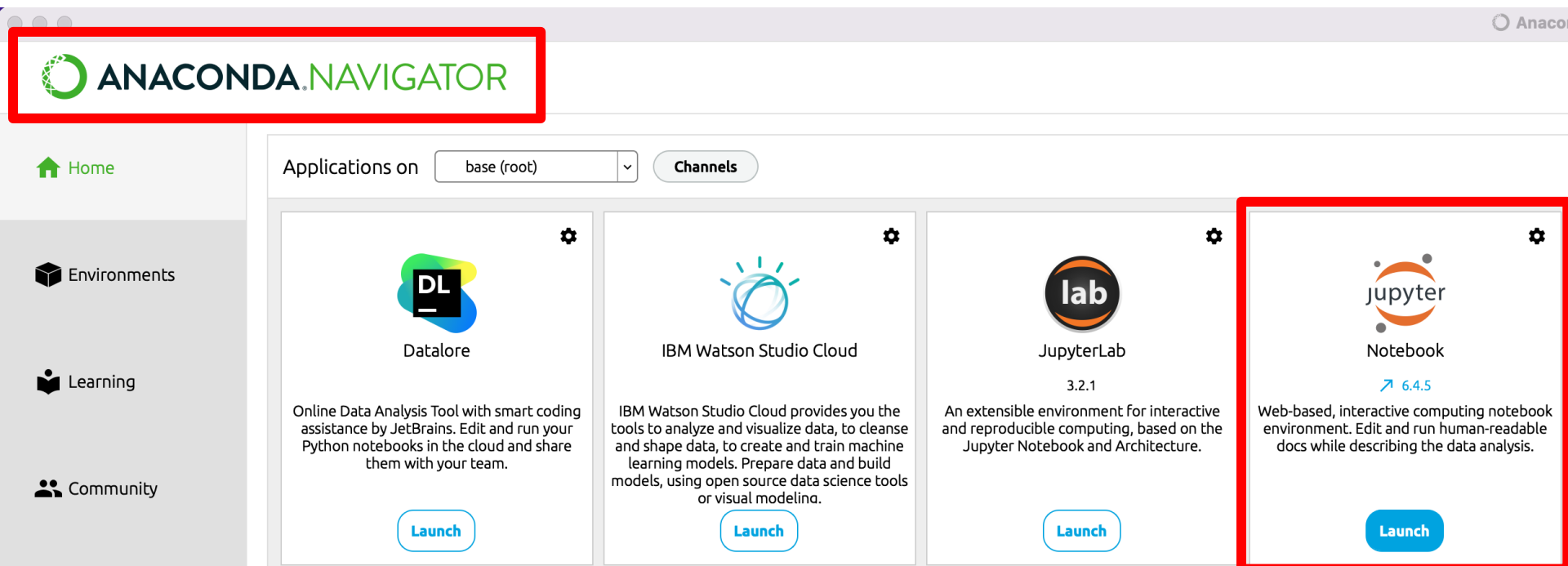
64-Bit (x86) Installer (581 MB)

64-Bit (Power8 and Power9) Installer (255 MB)

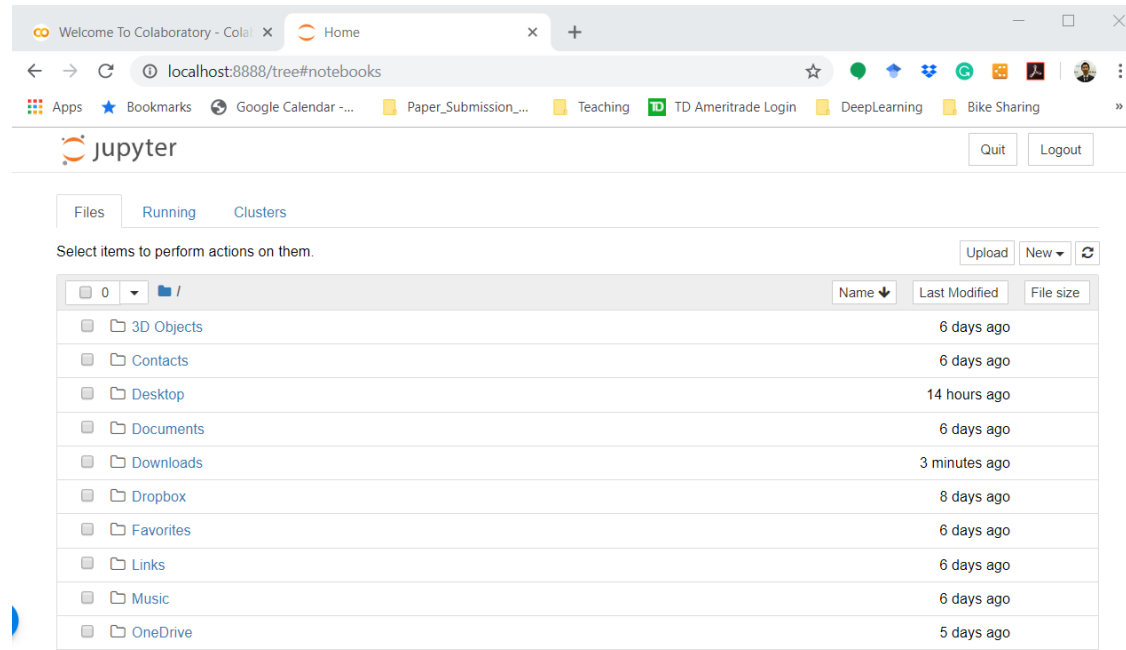
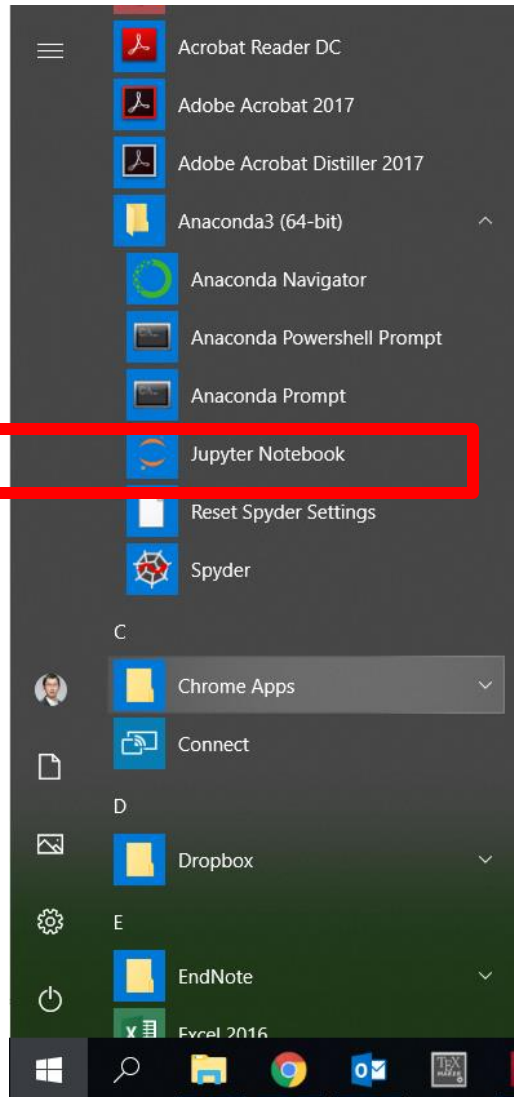
64-Bit (AWS Graviton2 / ARM64) Installer (488 M)

64-bit (Linux on IBM Z & LinuxONE) Installer (242 M)

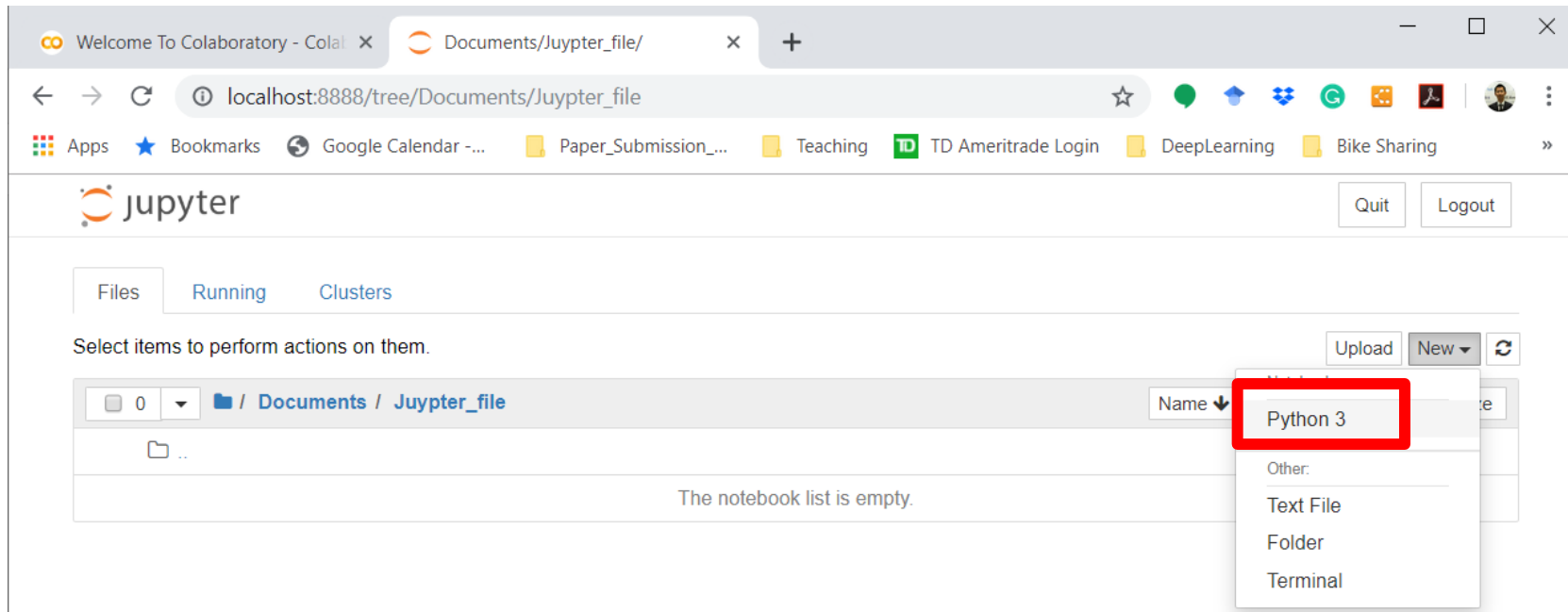
Anaconda – Jupyter notebook



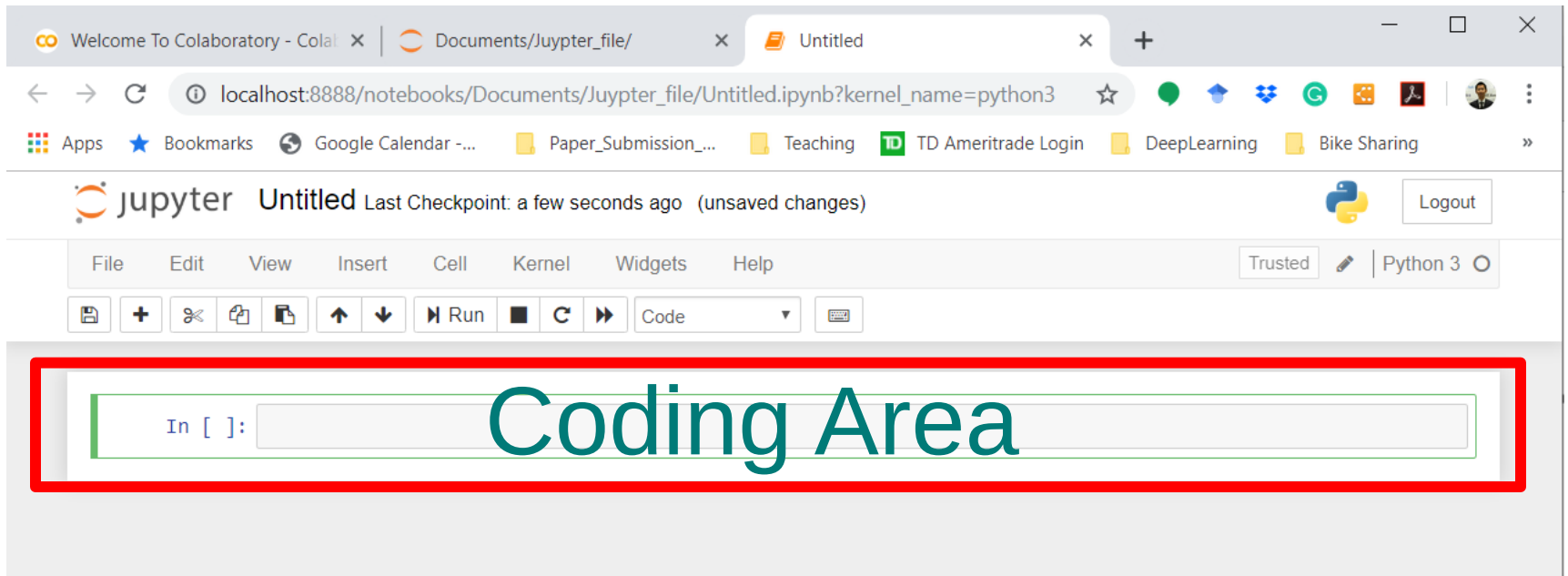
Jupyter notebook



Jupyter notebook



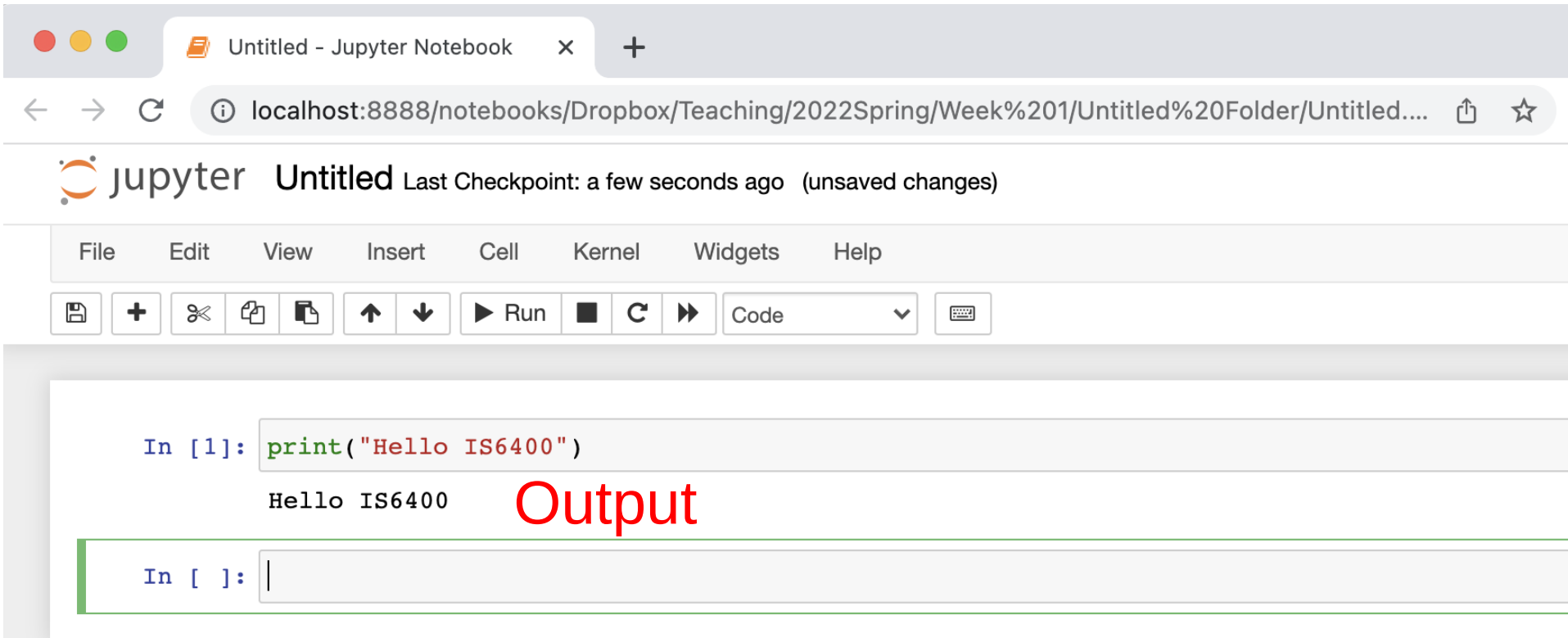
Jupyter notebook



Jupyter notebook

The screenshot displays a web browser window with a Jupyter Notebook. The browser's address bar shows the URL: `localhost:8888/notebooks/Dropbox/Teaching/2022Spring/Week%201/Untitled%20Folder/Untitled...`. The Jupyter interface includes a menu bar with options: File, Edit, View, Insert, Cell, Kernel, Widgets, and Help. Below the menu is a toolbar with icons for saving, adding, deleting, and running code. The 'Run' button, represented by a black square icon, is highlighted with a red arrow. The main area shows a code cell with the input `In [1]: print("Hello IS6400")` and the output `Hello IS6400`. A second red arrow points from the text **Type and Run (shift + enter)** to the 'Run' button. Below the first cell is an empty code cell with the prompt `In [ ]: |`.

Jupyter notebook



The screenshot shows a web browser window with a single tab titled "Untitled - Jupyter Notebook". The address bar displays the URL: `localhost:8888/notebooks/Dropbox/Teaching/2022Spring/Week%201/Untitled%20Folder/Untitled....`. The Jupyter interface includes a header with the "jupyter" logo, the title "Untitled", and a status message "Last Checkpoint: a few seconds ago (unsaved changes)". Below the header is a menu bar with options: File, Edit, View, Insert, Cell, Kernel, Widgets, and Help. A toolbar contains icons for saving, adding new files, undo, redo, and running code. The main area shows a code cell with the input `In [1]: print("Hello IS6400")` and its output `Hello IS6400`. A large red word "Output" is overlaid on the right side of the output text. Below the first cell is an empty code cell with the input `In [ ]: |`.

File Edit View Insert Cell Kernel Widgets Help

Save + Undo Redo Run Stop Restart Code

```
In [1]: print("Hello IS6400")
```

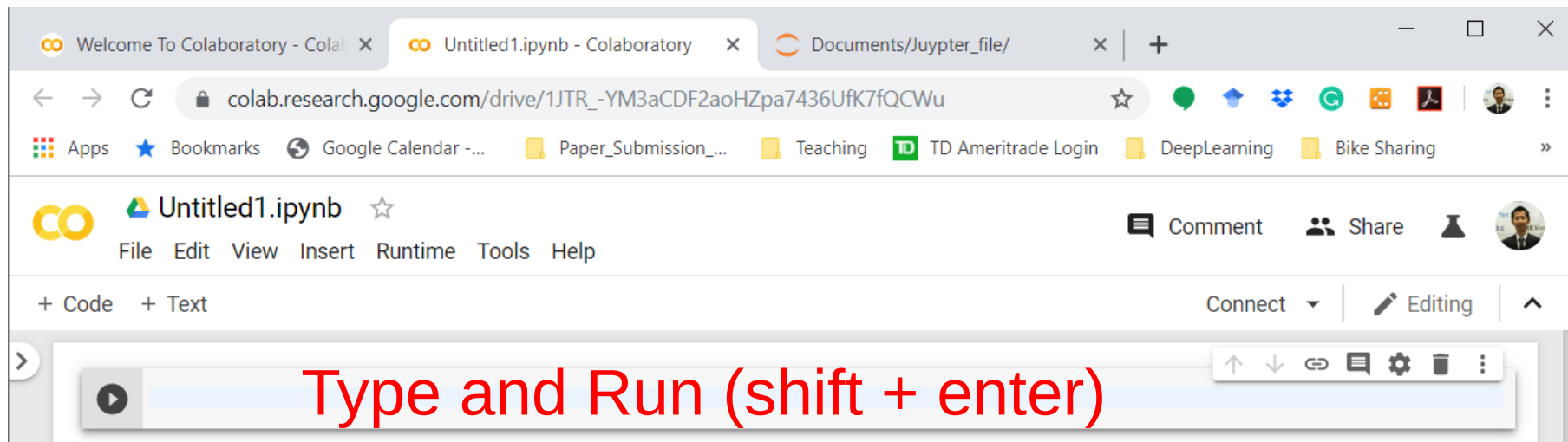
Hello IS6400

Output

```
In [ ]: |
```

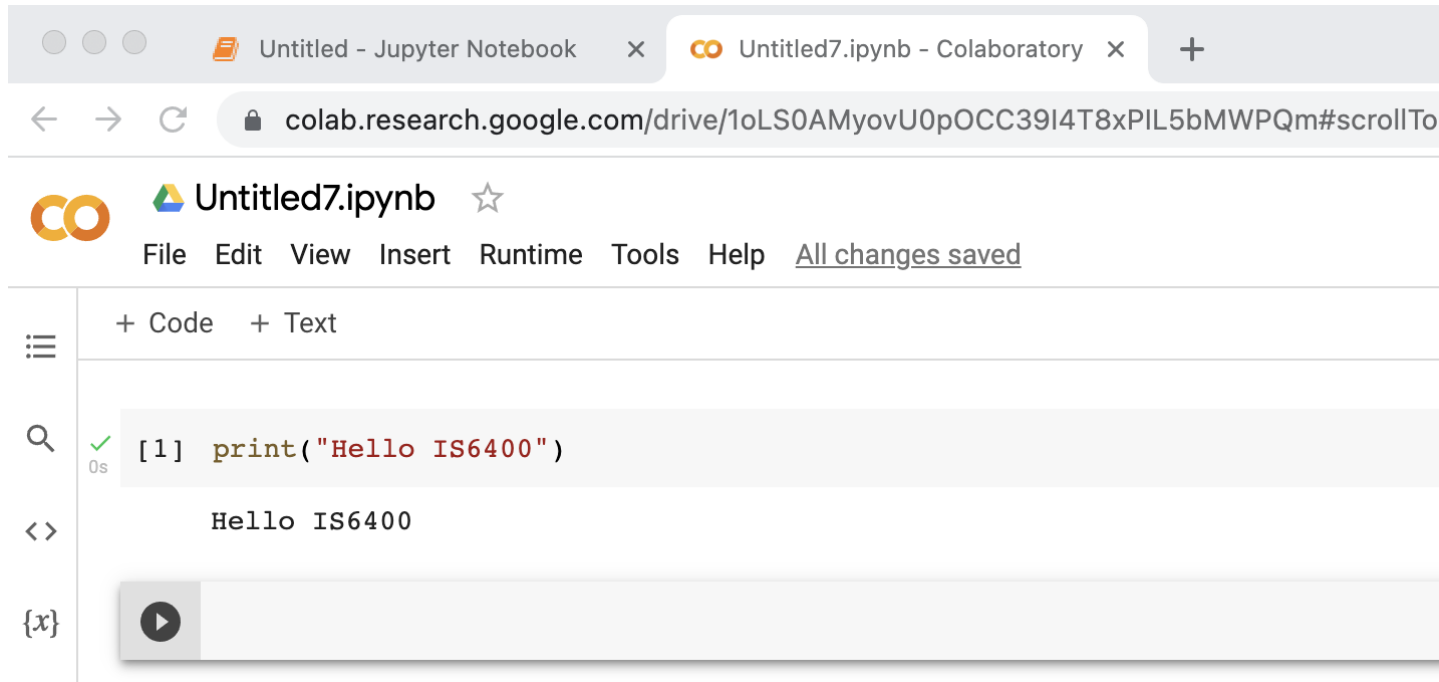
Google Colab

❖ <https://colab.research.google.com/>

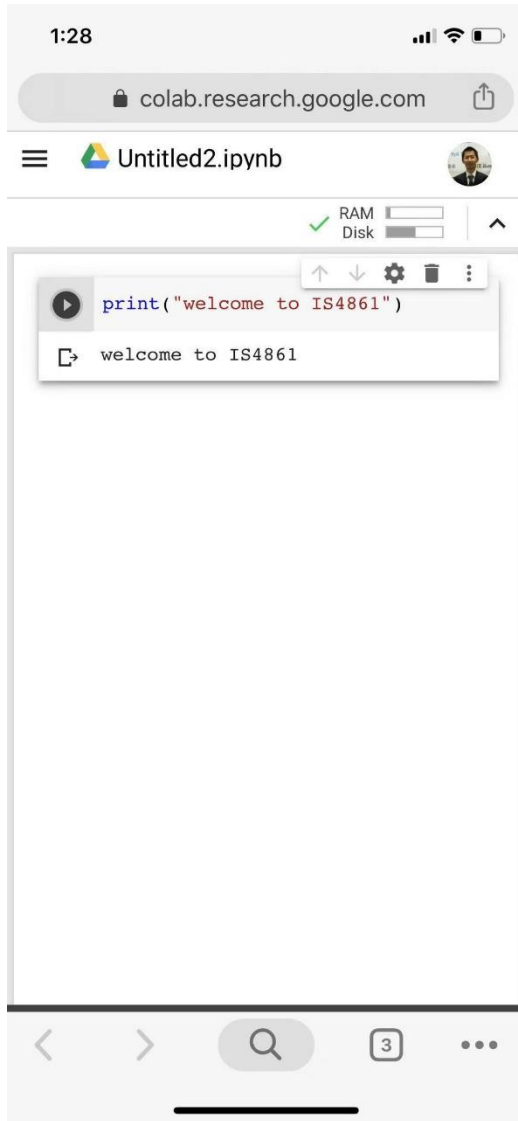


Google Colab

❖ <https://colab.research.google.com/>



Google Colab



Use my cell phone to code?
YES!